Please check the examination details below before entering your candidate information

Candidate surname	Other names

Centre Number | Candidate Number

Pearson Edexcel Level 3 GCE

Friday 16 May 2025

Afternoon

Paper reference **8FM0/21**

Further Mathematics

Advanced Subsidiary
Further Mathematics options
21: Further Pure Mathematics 1
(Part of options A, B, C and D)

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 5 questions.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

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Turn over

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Pearson

1. **In this question you must show all stages of your working.**
Solutions based entirely on calculator technology are not acceptable.

The surface temperature of the water in a lake during a particular year is modelled by the equation

$$S = 12 - \frac{15}{2}\cos x° - \frac{27}{10}\sin x° \qquad (\text{I})$$

where S is the temperature in degrees Celsius and x is the number of days after the start of the year.

(a) Use the model to write down the surface temperature of the water in the lake at the start of the year.

(1)

Using the substitution $t = \tan\left(\frac{x}{2}\right)$

(b) show that equation (I) can be rewritten as

$$S = \frac{At^2 + Bt + C}{10(1 + t^2)}$$

where A, B and C are integers to be determined.

(3)

(c) Hence determine, according to the model, the number of days after the start of the year when the surface temperature of the water in the lake is 10 °C for the **second** time that year. Give your answer to the nearest day.

(5)

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Question 1 continued

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Question 1 continued

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Question 1 continued

(Total for Question 1 is 9 marks)

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2. Water is leaking from a hole in the base of a large spherical tank. The depth of water, H metres, in the tank is modelled by the differential equation

$$3(5H - H^2)\frac{\mathrm{d}H}{\mathrm{d}t} = -4\sqrt{H} \qquad 0 < H < 5$$

where t is the time in hours after the leak started.

The depth of water in the tank 10 minutes after the leak started was 2 m.

Use two applications of the approximation formula

$$\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)_n \approx \frac{(y_{n+1} - y_n)}{h}$$

to estimate the depth of water in the tank, one hour after the leak started.

(7)

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Question 2 continued

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Question 2 continued

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Question 2 continued

(Total for Question 2 is 7 marks)

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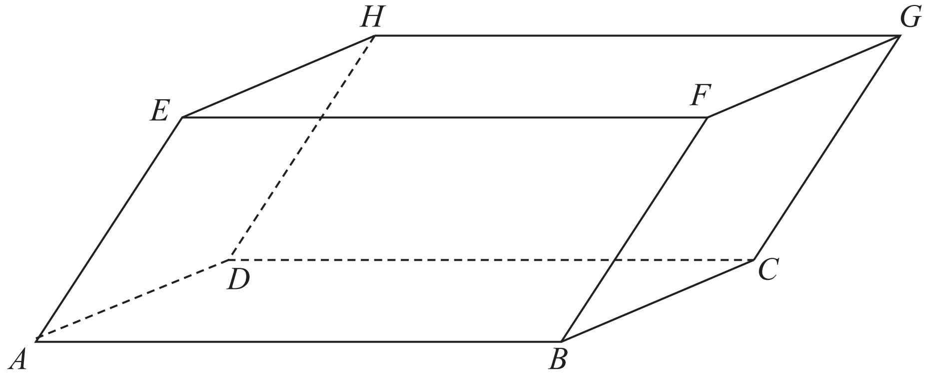

Figure 1

Figure 1 shows the parallelepiped *ABCDHEFG*.

Relative to a fixed origin O, the points A, B, D and E have coordinates (3, 2, 7), (4, 4, 3), (4, 1, t) and (t, –1, 10) respectively, where t is a constant.

(a) Determine, in simplest form in terms of t, $\overrightarrow{AB} \times \overrightarrow{AD}$

(3)

Given that the volume of the parallelepiped is 9

(b) determine the possible values of t.

(6)

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Question 3 continued

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Question 3 continued

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Question 3 continued

(Total for Question 3 is 9 marks)

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4. **In this question you must show all stages of your working.**
Solutions based on calculator technology are not acceptable.

Determine the values of x for which

$$\frac{x-8}{x} \leqslant \frac{7}{x(x-2)}$$

giving your answer in set notation.

(6)

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Question 4 continued

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Question 4 continued

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Question 4 continued

(Total for Question 4 is 6 marks)

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5. The parabola C has equation $y^2 = 16x$

The point $P(4t^2, 8t)$ lies on C.

(a) Show that an equation for the tangent to C at P is

$$yt - x = 4t^2$$

(3)

The line l passes through the origin and is perpendicular to the tangent to C at P.
The line l and the tangent to C at P intersect at the point Q.

(b) Determine, in simplest form in terms of t, the coordinates of Q.

(3)

(c) Hence show that, as t varies, the point Q lies on the curve with equation

$$y^2 = \frac{Ax^3}{x + B}$$

where A and B are integers to be determined.

(3)

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Question 5 continued

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Question 5 continued

(Total for Question 5 is 9 marks)

TOTAL FOR FURTHER PURE MATHEMATICS 1 IS 40 MARKS

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Please check the examination details below before entering your candidate information

Candidate surname	Other names

Centre Number | Candidate Number

Pearson Edexcel Level 3 GCE

Friday 17 May 2024

Afternoon

Paper reference **8FM0/21**

Further Mathematics

Advanced Subsidiary
Further Mathematics options
21: Further Pure Mathematics 1
(Part of options A, B, C and D)

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

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- Check your answers if you have time at the end.

Turn over ▶

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1. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

(a) Sketch the graph of the curve with equation

$$y = \frac{1}{x^2}$$

(2)

(b) Solve, using algebra, the inequality

$$3 - 2x^2 > \frac{1}{x^2}$$

(5)

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Question 1 continued

(Total for Question 1 is 7 marks)

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2. An area of woodland contains a mixture of blue and yellow flowers.

A study found that the proportion, x, of blue flowers in the woodland area satisfies the differential equation

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{xt(0.8 - x)}{x^2 + 5t} \qquad t > 0$$

where t is the number of years since the start of the study.

Given that exactly 3 years after the start of the study half of the flowers in the woodland area were blue,

(a) use one application of the approximation formula $\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)_n \approx \frac{y_{n+1} - y_n}{h}$ to estimate the proportion of blue flowers in the woodland area half a year later.

(5)

(b) Deduce from the differential equation the proportion of flowers that will be blue in the long term.

(1)

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Question 2 continued

(Total for Question 2 is 6 marks)

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3. Vectors $\mathbf{u}$ and $\mathbf{v}$ are given by

$$\mathbf{u} = 5\mathbf{i} + 4\mathbf{j} - 3\mathbf{k} \quad \text{and} \quad \mathbf{v} = a\mathbf{i} - 6\mathbf{j} + 2\mathbf{k}$$

where a is a constant.

(a) Determine, in terms of a, the vector product $\mathbf{u} \times \mathbf{v}$

(2)

Given that

- $\overrightarrow{AB} = 2\mathbf{u}$
- $\overrightarrow{AC} = \mathbf{v}$
- the area of triangle ABC is 15

(b) determine the possible values of a.

(4)

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Question 3 continued

(Total for Question 3 is 6 marks)

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4. (a) Given that $t = \tan\frac{x}{2}$ prove that

$$\cos x \equiv \frac{1-t^2}{1+t^2}$$

(3)

(b) Show that the equation

$$3\tan x - 10\cos x = 10$$

can be written in the form

$$(t+2)\left(at^2 + bt + c\right) = 0$$

where $t = \tan\frac{x}{2}$ and a, b and c are integers to be determined.

(4)

(c) Hence solve, for $-180° < x < 180°$, the equation

$$3\tan x - 10\cos x = 10$$

(5)

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Question 4 continued

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Question 4 continued

(Total for Question 4 is 12 marks)

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5. The parabola C has equation $y^2 = 16x$

The point P on C has y coordinate p, where p is a positive constant.

(a) Show that an equation of the tangent to C at P is given by

$$2py = 16x + p^2$$

$$\left[\text{You may quote without proof that for the general parabola } y^2 = 4ax,\ \frac{dy}{dx} = \frac{2a}{y}\right]$$

(2)

(b) Write down the equation of the directrix of C.

(1)

The line l is the reflection of the tangent to C at P in the directrix of C.

Given that l passes through the focus of C,

(c) determine the exact value of p.

(6)

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Question 5 continued

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Question 5 continued

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Question 5 continued

(Total for Question 5 is 9 marks)

TOTAL FOR FURTHER PURE MATHEMATICS 1 IS 40 MARKS

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Candidate surname	Other names

Centre Number | Candidate Number

Pearson Edexcel Level 3 GCE

Friday 19 May 2023

Afternoon

Paper reference **8FM0/21**

Further Mathematics

Advanced Subsidiary
Further Mathematics options
21: Further Pure Mathematics 1
(Part of options A, B, C and D)

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

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Information

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- The total mark for this part of the examination is 40. There are 6 questions.
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Advice

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- Try to answer every question.
- Check your answers if you have time at the end.

Turn over

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P 7 2 8 0 7 A 0 1 1 6

Pearson

1. (a) Use algebra to determine the values of x for which

$$\frac{5x}{x-2} \geqslant 12$$

(4)

(b) Hence, given that x is an integer, deduce the value of x.

(1)

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Question 1 continued

(Total for Question 1 is 5 marks)

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2. (a) Use the substitution $t = \tan\left(\frac{x}{2}\right)$ to show that the equation

$$3\cos x - 2\sin x = 1$$

can be written in the form

$$2t^2 + 2t - 1 = 0$$

(3)

(b) Hence solve, for $-180° < x < 180°$, the equation

$$3\cos x - 2\sin x = 1$$

giving your answers to one decimal place.

(4)

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Question 2 continued

(Total for Question 2 is 7 marks)

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3. The rectangular hyperbola H has equation $xy = c^2$ where c is a positive constant.

The line l has equation $x - 2y = c$

The points P and Q are the points of intersection of H and l

(a) Determine, in terms of c, the coordinates of P and the coordinates of Q

(3)

The point R is the midpoint of PQ

(b) Show that, as c varies, the coordinates of R satisfy the equation

$$xy = -\frac{c^2}{a}$$

where a is a constant to be determined.

(2)

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Question 3 continued

(Total for Question 3 is 5 marks)

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4. A teacher made a cup of coffee. The temperature θ°C of the coffee, t minutes after it was made, is modelled by the differential equation

$$\frac{\mathrm{d}\theta}{\mathrm{d}t} + 0.05(\theta - 20) = 0$$

Given that

- the initial temperature of the coffee was 95 °C
- the coffee can only be safely drunk when its temperature is below 70 °C
- the teacher made the cup of coffee at 1.15 pm
- the teacher needs to be able to start drinking the coffee by 1.20 pm

use two iterations of the approximation formula

$$\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)_n \approx \frac{y_{n+1} - y_n}{h}$$

to estimate whether the teacher will be able to start drinking the coffee at 1.20 pm.

(6)

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Question 4 continued

(Total for Question 4 is 6 marks)

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5. The points A, B and C are the vertices of a triangle.

Given that

- $\overrightarrow{AB} = \begin{pmatrix} p \\ 4 \\ 6 \end{pmatrix}$ and $\overrightarrow{AC} = \begin{pmatrix} q \\ 4 \\ 5 \end{pmatrix}$ where p and q are constants
- $\overrightarrow{AB} \times \overrightarrow{AC}$ is parallel to $2\mathbf{i} + 3\mathbf{j} + 4\mathbf{k}$

(a) determine the value of p and the value of q

(7)

(b) Hence, determine the exact area of triangle ABC

(2)

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Question 5 continued

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Question 5 continued

(Total for Question 5 is 9 marks)

Turn over

6. The parabola C has equation $y^2 = 4ax$ where a is a positive constant.

The point $P(at^2, 2at)$, $t \neq 0$, lies on C

The normal to C at P is parallel to the line with equation $y = 2x$

(a) For the point P, show that $t = -2$

(3)

The normal to C at P intersects C again when $x = 9$

(b) Determine the value of a, giving a reason for your answer.

(5)

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Question 6 continued

Question 6 continued

(Total for Question 6 is 8 marks)

TOTAL FOR FURTHER PURE MATHEMATICS 1 IS 40 MARKS

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Candidate surname	Other names

Centre Number ☐☐☐☐☐ Candidate Number ☐☐☐☐

Pearson Edexcel Level 3 GCE

Paper reference **8FM0/21**

Further Mathematics

Advanced Subsidiary
Further Mathematics options
21: Further Pure Mathematics 1
(Part of options A, B, C and D)

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Total Marks

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Turn over ▶

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Pearson

1. Use algebra to find the set of values of x for which

$$x \geqslant \frac{2x + 15}{2x + 3}$$

(6)

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Question 1 continued

(Total for Question 1 is 6 marks)

2. A population of deer was introduced onto an island.

The number of deer, P, on the island at time t years following their introduction is modelled by the differential equation

$$\frac{\mathrm{d}P}{\mathrm{d}t} = \frac{P}{5000}\left(1000 - \frac{P(t+1)}{6t+5}\right) \qquad t > 0$$

It was estimated that there were 540 deer on the island six months after they were introduced.

Use **two** applications of the approximation formula $\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)_n \approx \frac{y_{n+1} - y_n}{h}$ to estimate the number of deer on the island 10 months after they were introduced.

(7)

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Question 2 continued

(Total for Question 2 is 7 marks)

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Turn over

3. (a) Use $t = \tan\dfrac{\theta}{2}$ to show that, where both sides are defined

$$\frac{29 - 21\sec\theta}{20 - 21\tan\theta} \equiv \frac{5t + 2}{2t + 5}$$

(4)

(b) Hence, again using $t = \tan\dfrac{\theta}{2}$, prove that, where both sides are defined

$$\frac{20 + 21\tan\theta}{29 + 21\sec\theta} \equiv \frac{29 - 21\sec\theta}{20 - 21\tan\theta}$$

(3)

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Question 3 continued

(Total for Question 3 is 7 marks)

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4. The parabola C has equation $y^2 = 10x$

The point F is the focus of C.

(a) Write down the coordinates of F.

(1)

The point P on C has y coordinate q, where $q > 0$

(b) Show that an equation for the tangent to C at P is given by

$$10x - 2qy + q^2 = 0$$

(3)

The tangent to C at P intersects the directrix of C at the point A.

The point B lies on the directrix such that PB is parallel to the x-axis.

(c) Show that the point of intersection of the diagonals of quadrilateral $PBAF$ always lies on the y-axis.

(5)

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Question 4 continued

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Question 4 continued

(Total for Question 4 is 9 marks)

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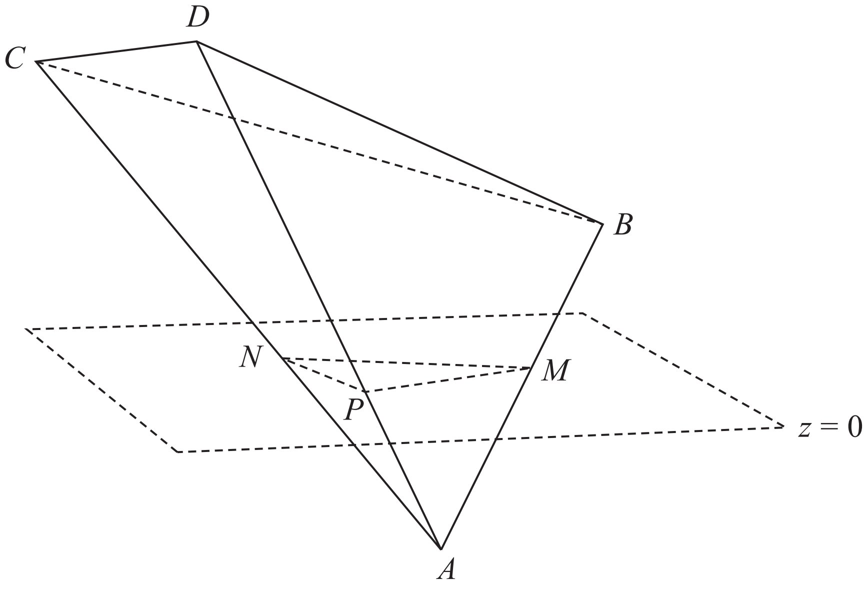

Figure 1

The points $A(3, 2, -4)$, $B(9, -4, 2)$, $C(-6, -10, 8)$ and $D(-4, -5, 10)$ are the vertices of a tetrahedron.

The plane with equation $z = 0$ cuts the tetrahedron into two pieces, one on each side of the plane.

The edges AB, AC and AD of the tetrahedron intersect the plane at the points M, N and P respectively, as shown in Figure 1.

Determine

(a) the coordinates of the points M, N and P,

(3)

(b) the area of triangle MNP,

(2)

(c) the exact volume of the solid $BCDPNM$.

(6)

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Question 5 continued

Question 5 continued

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Question 5 continued

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Question 5 continued

(Total for Question 5 is 11 marks)

TOTAL FOR FURTHER PURE MATHEMATICS 1 IS 40 MARKS

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Candidate surname	Other names

Pearson Edexcel Level 3 GCE

Centre Number | Candidate Number

Paper reference **8FM0/21**

Further Mathematics

Advanced Subsidiary
Further Mathematics options
21: Further Pure Mathematics 1
(Part of options A, B, C and D)

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Total Marks

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- Answer the questions in the spaces provided – *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 5 questions.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.
- Good luck with your examination.

Turn over ▶

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1. Use algebra to determine the values of x for which

$$x(x-1) > \frac{x-1}{x}$$

giving your answer in set notation.

(6)

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Question 1 continued

(Total for Question 1 is 6 marks)

2. The variables x and y satisfy the differential equation

$$\frac{d^2y}{dx^2} + 15\frac{dy}{dx} - 3y^2 = 2x$$

where $y = 1$ at $x = 0$ and where $y = 2$ at $x = 0.1$

Use the approximations

$$\left(\frac{d^2y}{dx^2}\right)_n \approx \frac{\left(y_{n+1} - 2y_n + y_{n-1}\right)}{h^2} \text{ and } \left(\frac{dy}{dx}\right)_n \approx \frac{\left(y_{n+1} - y_{n-1}\right)}{2h}$$

with $h = 0.1$ to find an estimate for the value of y when $x = 0.3$

(6)

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Question 2 continued

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Question 2 continued

(Total for Question 2 is 6 marks)

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3. On a particular day, the depth of water in a river estuary at a specific location is modelled by the equation

$$D = 2\sin\left(\frac{x}{3}\right) + 3\cos\left(\frac{x}{3}\right) + 6 \qquad 0 \leqslant x \leqslant 7\pi \qquad \text{(I)}$$

where the depth of water is D metres at time x hours after midnight on that day.

(a) Write down the depth of water at midnight, according to the model.

(1)

Using the substitution $t = \tan\left(\frac{x}{6}\right)$

(b) show that equation (I) can be re-written as

$$D = \frac{3t^2 + 4t + 9}{1 + t^2}$$

(3)

(c) Hence determine, according to the model, the time after midnight when the depth of water is 5 metres for the first time. Give your answer to the nearest minute.

(5)

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Question 3 continued

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Question 3 continued

(Total for Question 3 is 9 marks)

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4. With respect to a fixed origin O, the points A, B and C have position vectors given by

$$\overrightarrow{OA} = 18\mathbf{i} - 14\mathbf{j} - 2\mathbf{k} \quad \overrightarrow{OB} = -7\mathbf{i} - 5\mathbf{j} + 3\mathbf{k} \quad \overrightarrow{OC} = -2\mathbf{i} - 9\mathbf{j} - 6\mathbf{k}$$

The points O, A, B and C form the vertices of a tetrahedron.

(a) Show that the area of the triangular face ABC of the tetrahedron is 108 to 3 significant figures.

(3)

(b) Find the volume of the tetrahedron.

(4)

An oak wood block is made in the shape of the tetrahedron, with centimetres taken for the units.

The density of oak is $0.85\,\mathrm{g\,cm^{-3}}$

(c) Determine the mass of the block, giving your answer in kg.

(2)

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Question 4 continued

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Question 4 continued

(Total for Question 4 is 9 marks)

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5. The point $P(ap^2, 2ap)$, where a is a positive constant, lies on the parabola with equation

$$y^2 = 4ax$$

The normal to the parabola at P meets the parabola again at the point $Q(aq^2, 2aq)$

(a) Show that

$$q = \frac{-p^2 - 2}{p}$$

(5)

(b) Hence show that

$$PQ^2 = \frac{ka^2}{p^4}\left(p^2 + 1\right)^n$$

where k and n are integers to be determined.

(5)

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Question 5 continued

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Question 5 continued

(Total for Question 5 is 10 marks)

TOTAL FOR FURTHER PURE MATHEMATICS 1 IS 40 MARKS

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Please check the examination details below before entering your candidate information

Candidate surname	Other names

Pearson Edexcel Level 3 GCE

Centre Number

Candidate Number

Thursday 08 October 2020

Afternoon

Paper Reference **8FM0/21**

Further Mathematics

Advanced Subsidiary
Further Mathematics options
21: Further Pure Mathematics 1
(Part of options A, B, C and D)

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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Advice

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- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ▶

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1. The variables x and y satisfy the differential equation

$$\frac{d^2y}{dx^2} = 2y^2 - x - 1$$

where $\frac{dy}{dx} = 3$ and $y = 0$ at $x = 0$

Use the approximations

$$\left(\frac{d^2y}{dx^2}\right)_n \approx \frac{(y_{n+1} - 2y_n + y_{n-1})}{h^2} \quad \text{and} \quad \left(\frac{dy}{dx}\right)_n \approx \frac{(y_{n+1} - y_{n-1})}{2h}$$

with $h = 0.1$ to find an estimate for the value of y at $x = 0.2$

(7)

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Question 1 continued

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Question 1 continued

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Question 1 continued

(Total for Question 1 is 7 marks)

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2. Use algebra to determine the values of x for which

$$\frac{x+1}{2x^2+5x-3} > \frac{x}{4x^2-1}$$

(5)

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Question 2 continued

(Total for Question 2 is 5 marks)

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3. (i) Use the substitution $t = \tan\left(\frac{x}{2}\right)$ to prove that

$$\cot x + \tan\left(\frac{x}{2}\right) = \operatorname{cosec} x \qquad x \neq n\pi,\ n \in \mathbb{Z}$$

(2)

(ii)

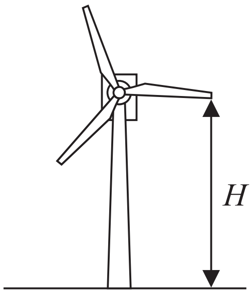

Figure 1

An engineer models the vertical height above the ground of the tip of one blade of a wind turbine, shown in Figure 1. The ground is assumed to be horizontal.

The vertical height of the tip of the blade above the ground, H metres, at time x seconds after the wind turbine has reached its constant operating speed, is modelled by the equation

$$H = 90 - 30\cos(120x)^\circ - 40\sin(120x)^\circ \qquad \text{(I)}$$

(a) Show that $H = 60$ when $x = 0$

(1)

Using the substitution $t = \tan(60x)^\circ$

(b) show that equation (I) can be rewritten as

$$H = \frac{120t^2 - 80t + 60}{1 + t^2}$$

(3)

(c) Hence find, according to the model, the value of x when the tip of the blade is 100 m above the ground for the first time after the wind turbine has reached its constant operating speed.

(5)

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Question 3 continued

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Question 3 continued

(Total for Question 3 is 11 marks)

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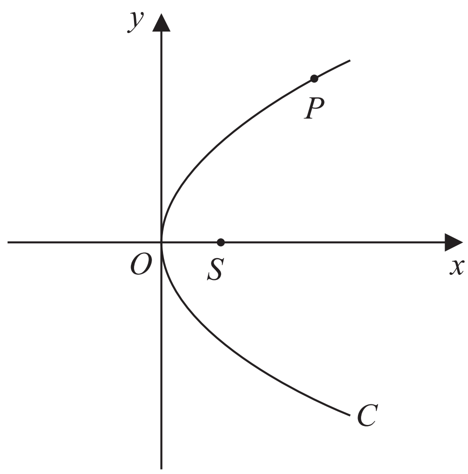

Figure 2

Figure 2 shows a sketch of the parabola C with equation $y^2 = 4ax$, where a is a positive constant. The point S is the focus of C and the point $P(ap^2, 2ap)$ lies on C where $p > 0$

(a) Write down the coordinates of S.

(1)

(b) Write down the length of SP in terms of a and p.

(1)

The point $Q(aq^2, 2aq)$, where $p \neq q$, also lies on C.
The point M is the midpoint of PQ.

Given that $pq = -1$

(c) prove that, as P varies, the locus of M has equation

$$y^2 = 2a(x - a)$$

(5)

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Question 4 continued

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Question 4 continued

(Total for Question 4 is 7 marks)

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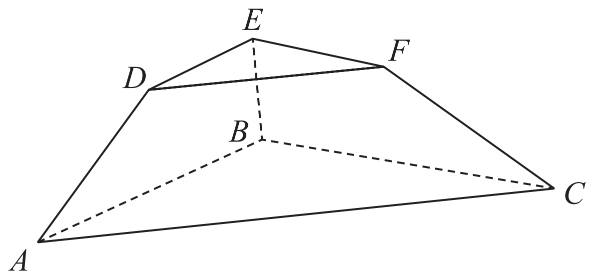

Figure 3

Figure 3 shows a solid display stand with parallel triangular faces ABC and DEF.
Triangle DEF is similar to triangle ABC.

With respect to a fixed origin O,
the points A, B and C have coordinates $(3, -3, 1)$, $(-5, 3, 3)$ and $(1, 7, 5)$ respectively and the points D, E and F have coordinates $(2, -1, 8)$, $(-2, 2, 9)$ and $(1, 4, 10)$ respectively.
The units are in centimetres.

(a) Show that the area of the triangular face DEF is $\frac{1}{2}\sqrt{339}\ \text{cm}^2$

(3)

(b) Find, in cm^3, the exact volume of the display stand.

(7)

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Question 5 continued

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Question 5 continued

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Question 5 continued

(Total for Question 5 is 10 marks)

TOTAL FOR FURTHER PURE MATHEMATICS 1 IS 40 MARKS

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Please check the examination details below before entering your candidate information

Candidate surname	Other names

Pearson Edexcel
Level 3 GCE

Centre Number | Candidate Number

Thursday 16 May 2019

Afternoon | Paper Reference **8FM0-21**

Further Mathematics

Advanced Subsidiary
Further Mathematics options
21: Further Pure Mathematics 1
(Part of options A, B, C and D)

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

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1. (a) Write down the t-formula for $\sin x$.

(1)

(b) Use the answer to part (a)

(i) to find the exact value of $\sin x$ when

$$\tan\left(\frac{x}{2}\right) = \sqrt{2}$$

(ii) to show that

$$\cos x = \frac{1-t^2}{1+t^2}$$

(4)

(c) Use the t-formulae to solve for $0 < \theta \leqslant 360°$

$$7\sin\theta + 9\cos\theta + 3 = 0$$

giving your answers to one decimal place.

(4)

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Question 1 continued

(Total for Question 1 is 9 marks)

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2. A student was set the following problem.

Use algebra to find the set of values of x for which

$$\frac{x}{x-24} > \frac{1}{x+11}$$

The student's attempt at a solution is written below.

$$x(x-24)(x+11)^2 > (x+11)(x-24)^2$$

$$x(x-24)(x+11)^2 - (x+11)(x-24)^2 > 0$$

$$(x-24)(x+11)\left[x(x+11) - x - 24\right] > 0 \qquad \text{Line 3}$$

$$(x-24)(x+11)\left[x^2 + 10x - 24\right] > 0$$

$$(x-24)(x+11)(x+12)(x-2) > 0$$

$$x = 24,\ x = -11,\ x = -12,\ x = 2$$

$$\{x \in \mathbb{R} : -12 < x < -11\} \cup \{x \in \mathbb{R} : 2 < x < 24\} \qquad \text{Line 7}$$

There are errors in the student's solution.

(a) Identify the error made

(i) in line 3

(ii) in line 7

(2)

(b) Find a correct solution to this problem.

(4)

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Question 2 continued

(Total for Question 2 is 6 marks)

Turn over

3. Julie decides to start a business breeding rabbits to sell as pets.

Initially she buys 20 rabbits. After t years the number of rabbits, R, is modelled by the differential equation

$$\frac{\mathrm{d}R}{\mathrm{d}t} = 2R + 4\sin t \qquad t > 0$$

Julie needs to have at least 40 rabbits before she can start to sell them.

Use two iterations of the approximation formula

$$\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)_n \approx \frac{y_{n+1} - y_n}{h}$$

to find out if, according to the model, Julie will be able to start selling rabbits after 4 months.

(7)

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Question 3 continued

(Total for Question 3 is 7 marks)

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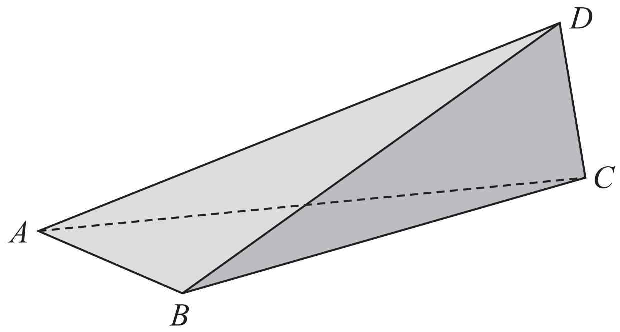

Figure 1

Figure 1 shows a sketch of a solid doorstop made of wood. The doorstop is modelled as a tetrahedron.

Relative to a fixed origin O, the vertices of the tetrahedron are A (2, 1, 4), B (6, 1, 2), C (4, 10, 3) and D (5, 8, d), where d is a positive constant and the units are in centimetres.

(a) Find the area of the triangle ABC.

(4)

Given that the volume of the doorstop is 21 cm^3

(b) find the value of the constant d.

(4)

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Question 4 continued

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Question 4 continued

(Total for Question 4 is 8 marks)

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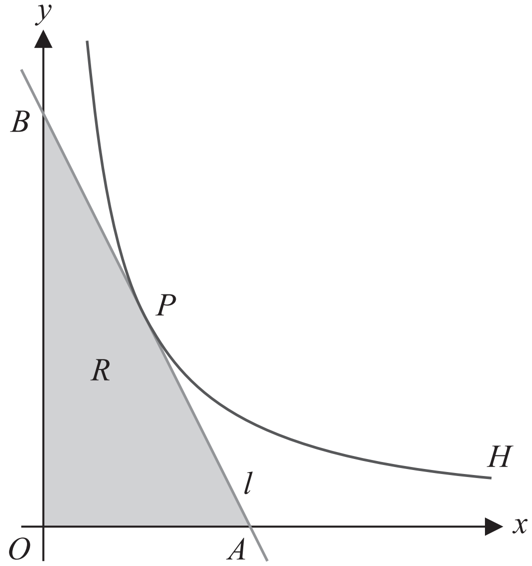

Figure 2

Figure 2 shows a sketch of part of the rectangular hyperbola H with equation

$$xy = c^2 \qquad x > 0$$

where c is a positive constant.

The point $P\left(ct, \frac{c}{t}\right)$ lies on H.

The line l is the tangent to H at the point P.

The line l crosses the x-axis at the point A and crosses the y-axis at the point B.

The region R, shown shaded in Figure 2, is bounded by the x-axis, the y-axis and the line l.

Given that the length OB is twice the length of OA, where O is the origin, and that the area of R is 32, find the exact coordinates of the point P.

(10)

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Question 5 continued

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Question 5 continued

(Total for Question 5 is 10 marks)

TOTAL FOR FURTHER PURE MATHEMATICS 1 IS 40 MARKS

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Write your name here

Surname	Other names

Pearson Edexcel
Level 3 GCE

Centre Number	Candidate Number

Further Mathematics

Advanced Subsidiary
Further Mathematics options
21: Further Pure Mathematics 1
(Part of options A, B, C and D)

Thursday 17 May 2018 – Afternoon

Paper Reference
8FM0/21

You must have:
Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator allowed by the regulations of the Joint Council for Qualifications. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

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Information

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Advice

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- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ▶

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Answer ALL questions. Write your answers in the spaces provided.

1. (a) Use the substitution $t = \tan\left(\frac{x}{2}\right)$ to show that the equation

$$5\sin x + 12\cos x = 2$$

can be written in the form

$$7t^2 - 5t - 5 = 0$$

(3)

(b) Hence solve, for $-180° < x < 180°$, the equation

$$5\sin x + 12\cos x = 2$$

giving your answers to one decimal place.

(4)

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Question 1 continued

(Total for Question 1 is 7 marks)

2. The temperature, θ °C, of coffee in a cup, t minutes after the cup of coffee is put in a room, is modelled by the differential equation

$$\frac{\mathrm{d}\theta}{\mathrm{d}t} = -k(\theta - 20)$$

where k is a constant.

The coffee has an initial temperature of 80 °C

Using $k = 0.1$

(a) use two iterations of the approximation formula $\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)_0 = \frac{y_1 - y_0}{h}$ to estimate the temperature of the coffee 3 minutes after it was put in the room.

(6)

The coffee in a different cup, which also had an initial temperature of 80 °C when it was put in the room, cools more slowly.

(b) Use this information to suggest how the value of k would need to be changed in the model.

(1)

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Question 2 continued

(Total for Question 2 is 7 marks)

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3. Use algebra to find the values of x for which

$$\frac{x}{x^2 - 2x - 3} \leqslant \frac{1}{x + 3}$$

(7)

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Question 3 continued

(Total for Question 3 is 7 marks)

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4. A scientist is investigating the properties of a crystal. The crystal is modelled as a tetrahedron whose vertices are $A(12, 4, -1)$, $B(10, 15, -3)$, $C(5, 8, 5)$ and $D(2, 2, -6)$, where the length of unit is the millimetre. The mass of the crystal is 0.5 grams.

(a) Show that, to one decimal place, the area of the triangular face ABC is $52.2\,\text{mm}^2$

(3)

(b) Find the density of the crystal, giving your answer in g cm^{-3}

(6)

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Question 4 continued

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Question 4 continued

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Question 4 continued

(Total for Question 4 is 9 marks)

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5. The rectangular hyperbola H has equation $xy = c^2$, where c is a non-zero constant.

The point $P\left(cp, \dfrac{c}{p}\right)$, where $p \neq 0$, lies on H.

(a) Use calculus to show that an equation of the normal to H at P is

$$p^3x - py + c(1 - p^4) = 0$$

(4)

The normal to H at the point P meets H again at the point Q.

(b) Find the coordinates of the midpoint of PQ in terms of c and p, simplifying your answer where possible.

(6)

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Question 5 continued

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Question 5 continued

(Total for Question 5 is 10 marks)

TOTAL FOR FURTHER PURE MATHEMATICS 1 IS 40 MARKS

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Write your name here

Surname	Other names

Pearson Edexcel Level 3 GCE

Centre Number	Candidate Number

Further Mathematics

Advanced Subsidiary
Further Mathematics options
Paper 2A: Further Pure Mathematics 1 and Further Pure Mathematics 2

Sample Assessment Material for first teaching September 2017
Time: 1 hour 40 minutes

Paper Reference
8FM0/2A

You must have:
Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- There are **two** sections in this question paper. Answer **all** the questions in Section A and **all** the questions in Section B.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 10 questions in this question paper. The total mark for this paper is 80.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ▶

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SECTION A

Answer ALL questions. Write your answers in the spaces provided.

1. (a) Use the substitution $t = \tan\left(\frac{x}{2}\right)$ to show that

$$\sec x - \tan x \equiv \frac{1-t}{1+t} \qquad x \neq (2n+1)\frac{\pi}{2},\ n \in \mathbb{Z}$$

(3)

(b) Use the substitution $t = \tan\left(\frac{x}{2}\right)$ and the answer to part (a) to prove that

$$\frac{1-\sin x}{1+\sin x} \equiv (\sec x - \tan x)^2 \qquad x \neq (2n+1)\frac{\pi}{2},\ n \in \mathbb{Z}$$

(3)

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Question 1 continued

(Total for Question 1 is 6 marks)

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2. The value, V hundred pounds, of a particular stock t hours after the opening of trading on a given day is modelled by the differential equation

$$\frac{\mathrm{d}V}{\mathrm{d}t} = \frac{V^2 - t}{t^2 + tV} \qquad 0 < t < 8.5$$

A trader purchases £300 of the stock one hour after the opening of trading.

Use two iterations of the approximation formula $\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)_0 \approx \frac{y_1 - y_0}{h}$ to estimate, to the nearest £, the value of the trader's stock half an hour after it was purchased.

(6)

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Question 2 continued

(Total for Question 2 is 6 marks)

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3. Use algebra to find the set of values of x for which

$$\frac{1}{x} < \frac{x}{x+2}$$

(6)

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Question 3 continued

(Total for Question 3 is 6 marks)

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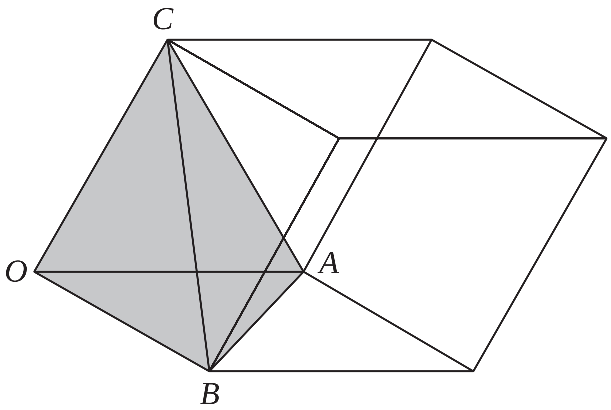

Figure 1

Figure 1 shows a sketch of a solid sculpture made of glass and concrete. The sculpture is modelled as a parallelepiped.

The sculpture is made up of a concrete solid in the shape of a tetrahedron, shown shaded in Figure 1, whose vertices are $O(0, 0, 0)$, $A(2, 0, 0)$, $B(0, 3, 1)$ and $C(1, 1, 2)$, where the units are in metres. The rest of the solid parallelepiped is made of glass which is glued to the concrete tetrahedron.

(a) Find the surface area of the glued face of the tetrahedron. **(4)**

(b) Find the volume of glass contained in this parallelepiped. **(5)**

(c) Give a reason why the volume of concrete predicted by this model may not be an accurate value for the volume of concrete that was used to make the sculpture. **(1)**

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Question 4 continued

(Total for Question 4 is 10 marks)

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5.

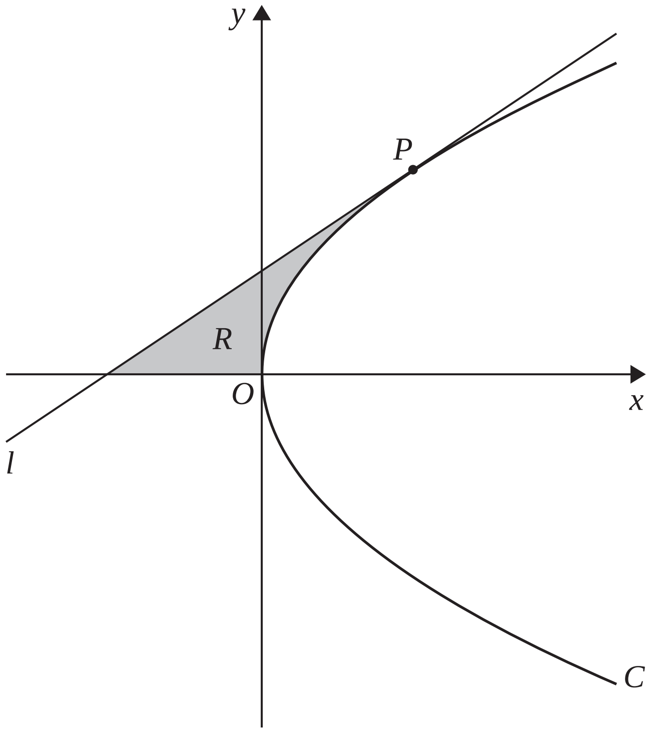

Diagram not drawn to scale

Figure 2

[You may quote without proof that for the general parabola $y^2 = 4ax$, $\dfrac{dy}{dx} = \dfrac{2a}{y}$]

The parabola C has equation $y^2 = 16x$.

(a) Deduce that the point $P(4p^2, 8p)$ is a general point on C.

(1)

The line l is the tangent to C at the point P.

(b) Show that an equation for l is

$$py = x + 4p^2$$

(3)

The finite region R, shown shaded in Figure 2, is bounded by the line l, the x-axis and the parabola C.

The line l intersects the directrix of C at the point B, where the y coordinate of B is $\dfrac{10}{3}$

Given that $p > 0$

(c) show that the area of R is 36

(8)

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Question 5 continued

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Question 5 continued

(Total for Question 5 is 12 marks)

TOTAL FOR SECTION A IS 40 MARKS

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Please check the examination details below before entering your candidate information

Candidate surname	Other names

Pearson Edexcel Level 3 GCE

Centre Number | Candidate Number

Specimen Paper

Paper Reference **8FM0/21**

Further Mathematics

Advanced Subsidiary

21: Further Pure Mathematics 1

You must have:
Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator allowed by the regulations of the Joint Council for Qualifications. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 5 questions in this question paper. The total mark for this paper is 40.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

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Answer ALL questions. Write your answers in the spaces provided.

1. The variable y satisfies the differential equation

$$\frac{d^2y}{dx^2} = 3 + 2y^2\frac{dy}{dx}$$

Given that $y = 1.5$ and $\frac{dy}{dx} = 2$ at $x = 1$, use the approximations

$$\left(\frac{d^2y}{dx^2}\right)_0 \approx \frac{y_1 - 2y_0 + y_{-1}}{h^2} \quad \text{and} \quad \left(\frac{dy}{dx}\right)_0 \approx \frac{y_1 - y_{-1}}{2h}$$

with $h = 0.1$, to find an estimate for y at $x = 0.9$

(5)

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Question 1 continued

(Total for Question 1 is 5 marks)

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2. Use algebra to find the exact set of values of x for which

$$\frac{3x}{2-x} \leqslant \frac{2}{x^2-4}$$

(7)

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Question 2 continued

(Total for Question 2 is 7 marks)

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3. The rectangular hyperbola H has Cartesian equation $xy = 15$

The distinct points $P\left(5p, \dfrac{3}{p}\right)$ and $Q\left(5q, \dfrac{3}{q}\right)$ lie on H, where $p \neq 0$ and $q \neq 0$

(a) Show that an equation of the tangent to H at P is

$$3x + 5p^2y = 30p$$

(4)

(b) Write down an equation of the tangent to H at Q.

(1)

The tangent to H at the point P and the tangent to H at the point Q intersect at the point R.

(c) Find, as single fractions in their simplest form, the coordinates of R in terms of p and q.

(4)

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Question 3 continued

(Total for Question 3 is 9 marks)

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4. With respect to the origin O, the points A, B, C and D have coordinates $(3, 6, 9)$, $(1, 5, 7)$, $(2, 3, 8)$ and $(3 + k, 6, 9 - k)$ respectively, where k is a non-zero constant.

(a) Show that $\overrightarrow{AD}$ is perpendicular to the plane ABC.

(2)

The right triangular prism with face ABC and edge AD has twice the volume of the tetrahedron $OBCD$.

(b) Find the possible values of k.

(7)

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Question 4 continued

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Question 4 continued

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Question 4 continued

(Total for Question 4 is 9 marks)

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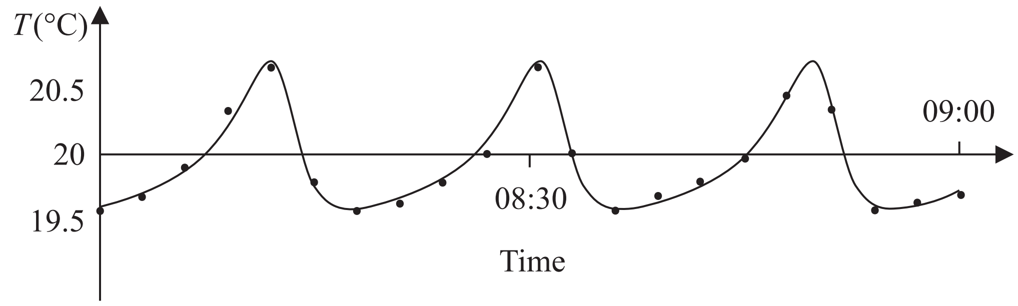

Figure 1

The temperature in a room of a house being regulated by a central heating system was recorded by an engineer every 3 minutes between 08:00 and 09:00 on a particular morning. The temperature outside at 08:00 was recorded as 15 °C.

Using radians, the engineer modelled the temperature, T °C, in the room x minutes after 08:00 by the equation

$$T = \frac{119 + 38\cos\left(\frac{x}{3}\right) + 79\sin\left(\frac{x}{3}\right)}{6 + 4\sin\left(\frac{x}{3}\right) + 2\cos\left(\frac{x}{3}\right)}$$

Figure 1 shows the recorded temperatures and the graph resulting from the engineer's model.

Using the t-substitution $t = \tan\left(\frac{x}{6}\right)$

(a) show that the equation can be rewritten as

$$T = \frac{81t^2 + 158t + 157}{4t^2 + 8t + 8}$$

(3)

The engineer assumes that while the heating system is switched on, the equation will continue to model the temperature beyond 09:00. Given that the heating system remains switched on,

(b) use the answer to part (a) to find the proportion of time that the temperature in the room will be above 20 °C according to the model.

(6)

(c) Give a reason why the equation may not be suitable to model the temperature in the room beyond 09:00.

(1)

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(Total for Question 5 is 10 marks)

TOTAL FOR PAPER IS 40 MARKS

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